Flash Evaporation and Modelling: 101

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1 The Process

- occurs when a sudden pressure drop causes the liquid to expand beyond its saturation conditions to a meta-stable state [1].
- observed when the surrounding liquid conditions suddenly change and become lower than its saturated conditions. Due to the sudden pressure drop, the whole energy cannot be contained in the liquid as sensisble heat and the heat surplus is converted into latent heat of vaporisation [2]
- the phase change can be triggered by depressurisation under nearly adiabatic conditions. In case of hot liquids, the temperature remains almost constant till the phase change starts, but the saturation temperature drops with the pressure. As the liquid pressure reaches the saturation pressure at the corresponding temperature, boiling (formation of vapour bubbles) takes place due to the interfacial heat transfer. The bubble growth and vapour generation is controlled by the heat transfer rate at the liquid-vapor interface. The pressure undershoot (difference between saturation and pressure at boiling inception) characterises the maximal non-equilibrium between the phases - its value depending on the initial liquid temperature, depressurisation rate, available nucleation sites, and the liquid and vapour properties [3]

1.1 'Flashing Inception'

- starts when the temperature increases by a few degrees above the saturation temperature or an initirally subcooled liquid may become saturated if the pressure drop is large enough to reduce the substance pressure below the saturation pressure corresponding to its temperature [4]
- occurs when the superheat limit is reached [5]
- themodynamic limit is given by $\frac{\partial P}{\partial V}|_T = 0$ along with an equation of state that is not available really. It's the spinodal curves that show this [5]
- in practice, the metastable liquid will undergo phase change before reaching the homogenous nucleation limit [5]

1.2 Nucleation

- as a significantly high local superheat degree does not exist, the location where nucleation is initiated is determined majorly by the available nucleation sites. In most cases, the walls are a major source of nucleation sites due to roughness and manufacturing imperfectness [6]

- in a superheated liquid, fluctuations of the fluid properties on a molecular level may cause the formation of vapour nuclei. The generation of vapour nuclei can be divided into homogenous nucleation within the liquid and heterogenous nucleation at the interface between the superheated liquid and a second (typically solid) phase [1].

- in most practical circumstances, the metastable liquid will undergo phase change before it reaches the homogenous nucleation limit. In this case, nucleation occurs around pre-existing nuclei or gaseous seeds or poorly wettable cavities on solid surfaces. This is known as heterogenous nucleation [5]

- — Zone I (subcooled liquid) — Zone II (metastable superheated liquid) — Zone III (nucleation; non-equilibrium) — Zone IV (equilibrium) —

- modelling methods
  - homogenous seeding (ignores zone II and III)
  - step function (ignores zone II)
  - nucleation modelling

1.3 Bubble Growth

- at the very early stage, the nuclei are small and surface tension is dominant and restricts growth, this delay period has a marginal influence and is often ignored [7, 6]

- second stage, called inertia controlled growth [1], starts after the nuclei have grown to around twice their diameter due to the dominant inertia forces. The radius of the nucleus bubble growth is a consequence of the pressure difference between the vapour pressure in the bubble and the exterior pressure. This is hydro-dynamically controlled [6]. Bubble growth is a linear function of time [7].

- the third stage is called the thermally controlled growth mode, where the after bubble grows further and its wall temperature decreases, the increase in heat transfer between the surrounding liquid and the bubble wall causes addition of vapour to the bubble by evaporation at the interface. Bubble growth is characterised by $R \propto t^{1/2}$ [7]. Growth is restricted mainly by the rate at which heat is supplied by the liquid. However, conduction is not always the primary means of heat transfer because turbulence and relative motion between bubbles and the surrounding liquid [5]

1.4 Vapour Generation Rate

Null.

2 Modelling Multiphase Flows

Two fundamentally different formulations fo the macroscopic field equations:

- Two Fluid Model: considers each phase separately; two sets of conservation equations of mass, momentum, and energy; coupling terms using the jump conditions.

- Drif Flux (Mixture) Model: consider mixture as a whole; three mixture conservation equations in addition to one diffusion equation (continuity) equation which takes into account the concentration changes - a mixture conservation equation is obtained by adding two conservation equations with an appropriate jump condition [8].

Several different models and simulation approaches for various two-phase flow conditions have evolved:

- Euler-Euler: Treats the two phase flow as interpenetrating continua. The governing equations of the fluid are solved for each phase on the Eulerian grid, and the interactions of the two phases at the interface are represented through suitable exchange terms.

- Euler-Lagrange: Models treat one phase as a continuous solution on an Eulerian grid. The second phase is dispersed and treated as discrete particles in a Lagrangian framework. Here, it is assumed that the bubbles, droplets, or solids represented by the particles can be treated as rigid bodies with homogenous properties within the partile and that they are in a dilute flow. The effect of the flow field around the particles is them subsumed as appropriate forces [1]

- Multiphase Mixture: Simplified Euler-Euler approach.

Among different approaches, the homogenous equilibrium model is generally recognised as the simplest. Within this approach, the flashing flow is considered as pseudo single-phase flowing through the nozzle at the equilibrium state. The HEM approach can be applied to relatively long nozzles (that allow for a smooth phase change inside them), however, in short nozzles, this can result in large errors as the equilibrium assumption is no longer valid. To accomodate the nonequilibrium effects, empirical correction factors, slip between phases (mixture models), and two fluid models are used. The needed closure models (flow pattern details, information about bubble size distributions and bubble shapes) are not general and are selected on a case-by-case basis. Approaches to modelling flashing flows need to consider: (a) the change of phase during flashing flow and (b) the interaction between the liquid phase and the vapor phase during the vaporization process. [9]

Concering the change of phase during flashing flow, the models can be classified into two main categories:

- models accounting for the nucleation process (which show better agreement with the real behaviour but require experimental data to tune the nucleation source term)

- models neglecting nucleation and assuming constant bubble size or number density

2.1 Local Instaneous Balance Equations

2.2 Averaging Techniques

[10, 11] [12, 13]

2.3 Volume Averages

All averaging procedures yield formally the same structure [11]

2.3.1 Prerequisites

- Phase Indicator Function: Given domains $\Omega_k(t)$, where $k = 1, 2$.
  - $$\Omega = \Omega_1(t) + \Omega_2(t) \tag{1}$$
    is constant.
  - interface separating the fluids is, $S_i(t) = (\partial\Omega_1 \bigcup \partial\Omega_2)/\partial\Omega$
  - phase indicator or characteristic function,
    $$\gamma_k(\mathbf{x}, t) = 1, if x \in \Omega_k(t) \tag{2}$$
  - $\nabla\gamma_k = \hat{\mathbf{n}}_\mathbf{k}\delta(\mathbf{x} - \mathbf{x}_i, t)$
  - Topological equation which expresses the fact that the material derivative of γ_k is zero,
    $$\frac{\partial\gamma_k}{\partial t} + \mathbf{v}_i \cdot \nabla\gamma_k = 0 \tag{3}$$
- Volume Averaging Operators: consider a general representative volume element V, bounded by ∂V, and invariant with respect to time and space. The centroid is at $\mathbf{x}$, with a relative position vector η
  $\implies \mathbf{x}_\eta = \mathbf{x} + \eta$
  - $\therefore$ Volume Fraction of Phase k,
    $$g_k = \frac{1}{V}\int_V \gamma_k(\mathbf{x} + \eta, t) d\mathbf{x}_\eta = \frac{V_k}{V} \tag{4}$$
  - for a general scalar or vector quantity. ψ_k, over the entire volume element V and the volume V_k occupied solely by the phase k,
    $$\overline{\psi_k}^V = \frac{1}{V}\int_V \psi_k(\mathbf{x} + \eta, t) X_k(\mathbf{x} + \eta, t) d\mathbf{x}_\eta, \tag{5}$$
    $$\overline{\psi_k}^k = \frac{1}{V_k}\int_V \psi_k(\mathbf{x} + \eta, t) X_k(\mathbf{x} + \eta, t) d\mathbf{x}_\eta \tag{6}$$
  - these averages are linearly related as, $\overline{\psi_k}^V = g_k\overline{\psi_k}^k$
- Gauß and Leibnitz Rules
- Fluctuation Component
- Theorems

2.4 Average macroscopic transport equations

Applying the general microscopic transport equation to phase k,

$$\frac{\partial\Phi_k}{\partial t} + \nabla \cdot (\Phi_k \mathbf{V}_k) = \nabla \cdot \mathbf{J}_k + Sk. \tag{7}$$

Multiplying by γ_k and averaging over dV,

$$\frac{1}{dV}\int_{dV} \frac{\partial\Phi_k}{\partial t}\gamma_k \, dv + \frac{1}{dV}\int_{dV} \nabla\cdot(\Phi_k \mathbf{V}_k)\gamma_k \, dv = \frac{1}{dV}\int_{dV} \nabla\cdot\mathbf{J}_k\gamma_k \, dv + \frac{1}{dV}\int_{dV} S_k\gamma_k \, dv. \tag{8}$$

Using Theorems 1 and 2, we get the exact form of the macroscopic transport equations [10],

$$\frac{\partial \Phi_k}{\partial t} + \nabla \cdot g_k \overline{\Phi_k}^k \overline{\mathbf{V}_k}^k = \nabla \cdot \overline{\mathbf{J}_k} + \overline{S_k} + \underbrace{\nabla \cdot \frac{1}{dV} \int_{dV} (-\hat{\Phi}_k \hat{\mathbf{V}}_k)\, dv}_{I_k^D} + \underbrace{\frac{1}{dV} \int_{dAk} \mathbf{J}_k \cdot \mathbf{n}_k \, da}_{I_k^J} + \underbrace{\frac{1}{dV} \int_{dA_k} \Phi_k (\mathbf{w}_k - \mathbf{V}_k) \cdot \mathbf{n}_k \, da}_{I_k^Q}. \quad (9)$$

Here, I_k^D is the microscopic deviation term and I_k^J and I_k^Q are the interfacial terms.

2.4.1 Assumptions

- Two phases only
- Variations of material properties in dV_k are neglected, although they may globally vary,

$$\text{averaged density of the phase k}\overline{\rho_k}^k = \rho_k = \overline{\rho_k}/g_k, \quad (10)$$

$$\overline{\rho_k \Phi_k} = \rho_k \overline{\Phi_k}, \quad (11)$$

$$\overline{\mu_k}^k = \mu_k, \quad (12)$$

$$\overline{k_k}^k = k_k, \quad (13)$$

$$\overline{D_k}^k = D_k. \quad (14)$$

2.4.2 Mass Conservation

$\Phi = 0$, $\mathbf{J} = \mathbf{0}$, and $S = 0$, thus, using equation 9,

$$\frac{\partial \overline{\rho_k}}{\partial t} + \nabla \cdot g_k \rho_k \overline{\mathbf{V_k}}^k = I_k^Q, \quad (15)$$

where $I_k^Q = \frac{1}{dV} \int_{dA_k} \Phi_k (\mathbf{w}_k - \mathbf{V}_k) \cdot \mathbf{n}_k \, da$. The microscopic deviation term is zero due to assumption 2. Considering the interfacial mass flux balance, $I_1^Q + I_2^Q = 0$. Thus,

$$\sum_{k=1}^{2} \frac{\partial \overline{\rho_k}}{\partial t} + \sum_{k=1}^{2} \nabla \cdot g_k \rho_k \overline{\mathbf{V}_k}^k = 0. \quad (16)$$

As $\overline{\rho_k} = \rho_k g_k$, and substituting angle brackets for overlines and using u instead of V ,

$$\frac{\partial}{\partial t}[\langle \rho_k \rangle g_k] + \nabla \cdot [g_k \langle \rho_k \mathbf{u}_k \rangle] = \Gamma_k, \quad (17)$$

where

$$\Gamma = -\frac{1}{V} \int_{S_i} \rho_k{}^1 (\mathbf{u}_k - \mathbf{u}_{\text{interface}}) \cdot \mathbf{n}_k \, dS_i, \quad (18)$$

and

$$\sum_{k=1}^{2} \Gamma_k = 0. \quad (19)$$

[1] this is not the averaged value

The convective mass flux of the phase across the interface is [11],

$$\dot{m}_k = \rho_k(\mathbf{u}_k - \mathbf{u}_{\text{interface}}) \cdot \mathbf{n}_k. \tag{20}$$

Written in one dimension, equation 17 can be written [14] as,

$$\frac{\partial}{\partial t}[\langle \rho_k \rangle g_k] + \frac{1}{A}\frac{\partial}{\partial z}[g_k \langle \rho_k u_k \rangle A] = \Gamma_k \tag{21}$$

And in cylindrical coordinates, this can be rewritten as [14, 8],

$$\frac{\partial}{\partial t}[\langle \rho_k \rangle g_k] + \frac{1}{r}\frac{\partial}{\partial r}[r g_k \langle \rho_k u_{rk} \rangle] + \frac{1}{r}\frac{\partial}{\partial \theta}[g_k \langle \rho_k u_{\theta k} \rangle] + \frac{\partial}{\partial z}[g_k \langle \rho_k u_{zk} \rangle] = \Gamma_k. \tag{22}$$

If flow is axisymmetric, there are no dependencies on θ and thus,

$$\frac{\partial}{\partial t}[r\langle \rho_k \rangle g_k] + \frac{1}{r}\frac{\partial}{\partial r}[r g_k \langle \rho_k u_{rk} \rangle] + \frac{\partial}{\partial z} r[g_k \langle \rho_k u_{zk} \rangle] = r\Gamma_k. \tag{23}$$

to solve the $\langle \rho_k \mathbf{u}_k \rangle$ issue, the weighted velocity (Favre average) can be taken so that $\langle \rho_k \mathbf{u}_k \rangle = \hat{\mathbf{u}}_k \langle \rho_k \rangle$.

2.4.3 Momentum Conservation

$\Phi = \rho \mathbf{u}$ and $S = \mathbf{b}$, where $\mathbf{b}$ is the body force.

For a Newtonian fluid flow, $\mathbf{J} + -\nabla P + \mu[\nabla \mathbf{u} + (\nabla \mathbf{u})^T]$. Thus, the averaged momentum equation yields,

$$\frac{\partial}{\partial t}[\rho \overline{\mathbf{u}_k}] + \nabla \cdot g_k \rho_k \overline{\mathbf{u}_k}^k \overline{\mathbf{u}_k}^k = -\nabla \overline{P_k} + \nabla \cdot [\overline{\mu_k \nabla \mathbf{u}_k} + \overline{\mu_k (\nabla \mathbf{u}_k)^T}] + I_k^D + I_k^Q + I_k^J + \overline{\mathbf{b}_k} \tag{24}$$

In a similar formulation,

$$\frac{\partial}{\partial t}[g_k \langle \rho_k \mathbf{u}_k \rangle] + \nabla \cdot [g_k \langle \rho_k \mathbf{u}_k \mathbf{u}_k \rangle] = -\nabla(g_k \langle p_k \rangle) + \nabla \cdot [g_k \langle \mathbb{T}_k \rangle] + g_k \langle \rho_k \mathbf{g} \rangle + \mathbf{M}_k \tag{25}$$

$\mathbf{M_k}$ represents the momentum transfer across the interface,

$$\mathbf{M}_k = -\overline{(-p_k \mathbb{I} + \mathbb{T}_k) \cdot \nabla X_k}^V, \tag{26}$$

along with the jump,

$$\sum_{k=1}^{2} \mathbf{M}_k - \mathbf{M}_m = 0 \tag{27}$$

where $\mathbf{M}_m$ is the mixture source. In one dimension, the momentum conservation for phase-k is [14],

$$\frac{\partial}{\partial t}[g_k \langle \rho_k u_k \rangle] + \frac{1}{A}\frac{\partial}{\partial z}[g_k \langle \rho_k u_k u_k \rangle A] = -g_k \frac{\partial p}{\partial z} + g \langle \rho_k \rangle g_k \cos\theta - \frac{1}{A}(P_{wk} u_{wk}) - \frac{1}{A}(P_i \tau_i) + \Gamma_k u_i. \tag{28}$$

The shear stresses acting on the phase at the wall and at the interface are denoted by τ_{wk} and τ_{ik} respectively. The part of wall perimeter wetted by phase k is P_{wk}, while P_i is the interfacial perimeter.

2.4.4 Energy Conservation

$\Phi = \rho H$, where H is the total enthalpy, and $S = 0$.

2.4.5 Interface Stuff

- At the interface, if there is phase change then the normal velocities at the interface are discontinious. The interface is no longer a material surface, since a mass transfer occures between the two phases.

2.4.6 Summary of Equations

In following the spirit of the subject, a complete change of notation. [MISSING CITATION!!]

- α_{jk} is the following fraction of field-j of phase-k.
- Γ_{jk} is the volumetric mass transfer rate due to phase change in field-j of phase-k
- m'''_{jk} is the mass source of field-j from other fields of phase-k
- $\underline{\underline{\tau}}^T_{jk}$ is the Reynolds stress for field-j of phase-k
- $\underline{\mathbf{M}}_{jk}, \underline{\mathbf{M}}^w_{jk}$ are the interfacial and wall forces (per unit volume)
- $\underline{\mathbf{v}}_i$ is the velocity of the interface
- $\mu = h - p/\rho$ (internal energy)
- $\underline{\mathbf{q}}''^T_{jk} = -\overline{h'_{jk}\underline{\mathbf{v}}'_{jk}}$ (turbulent heat flux)
- $q''_{i_{jk}} = \overline{\underline{\mathbf{q}}''_{jk} \cdot \nabla X_k}$ (interfacial heat flux)
- $\mathcal{D}_{jk} = \overline{X_k \underline{\underline{\tau}}_j : \nabla \underline{\mathbf{v}}_j}$ (dissipation)
- $p_{jk} = \overline{X_k p_j}$ (static pressure)

Mass conservation (field - j, phase - k):

$$\frac{\partial}{\partial t}(\alpha_{jk}\rho_k) + \nabla(\alpha_{jk}\rho_k\underline{\overline{\mathbf{v}}}_{jk}) = \Gamma_{jk} + m'''_{jk}. \tag{29}$$

Momentum conservation (field - j, phase - k):

$$\begin{aligned}\frac{\partial}{\partial t}(\alpha_{jk}\rho_k\underline{\overline{\mathbf{v}}}_{jk}) + \nabla\cdot(\alpha_{jk}\rho_k\underline{\overline{\mathbf{v}}}_{jk}\underline{\overline{\mathbf{v}}}_{jk}) = -\nabla(\alpha_{jk}p_{jk}) + \nabla\cdot(\alpha_{jk}[\underline{\underline{\overline{\tau}}}_{jk} + \underline{\underline{\overline{\tau}}}^T_{jk}]) \\ + \alpha_{jk}\rho_k\underline{\mathbf{g}} + \underline{\mathbf{M}}_{jk} + \underline{\mathbf{M}}^w_{jk} + \Gamma_{jk}\underline{\mathbf{v}}_i + m'''_{jk}\underline{\mathbf{v}}_{jk}.\end{aligned} \tag{30}$$

Energy conservation (field - j, phase - k):

$$\begin{aligned}\frac{\partial}{\partial t}(\alpha_{jk}\rho_k\overline{h}_{jk}) + \nabla\cdot(\alpha_{jk}\rho_k\underline{\overline{\mathbf{v}}}_{jk}\overline{h}_{jk}) = -\nabla\cdot\alpha_{jk}(\underline{\overline{\mathbf{q}''}}_{jk} + \underline{\mathbf{q}}''^T_{jk}) + \alpha_{jk}q'''_{jk} + \mathcal{D}_{jk} \\ + \frac{Dp_{jk}}{Dt} + q''_{i_{jk}}A'''_i + \Gamma_{jk}\mu_i + m'''_{jk}\mu_{jk}.\end{aligned} \tag{31}$$

2.5 Flow Models

2.6 Two Fluid Models

2.7 Drift Flux Models

The model consists of two continuity equations and a single combined momentum equation. The model may be completed with a combined energy equation or two energy equations. Wall effects are modelled using algebraic relations. [11]

2.7.1 Homogenous Flow Models

The assumption is made that the two phases move with the same velocity and correlation terms are usually ignored. The homogenous model therefore consits of the combined continuity and momentum equations, a rule to calculate the volume fractions and, possibly, a combined energy equation. [11]

The phases are in so called mechanical equilibrium, thus the relative velocities are zero. [13]

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